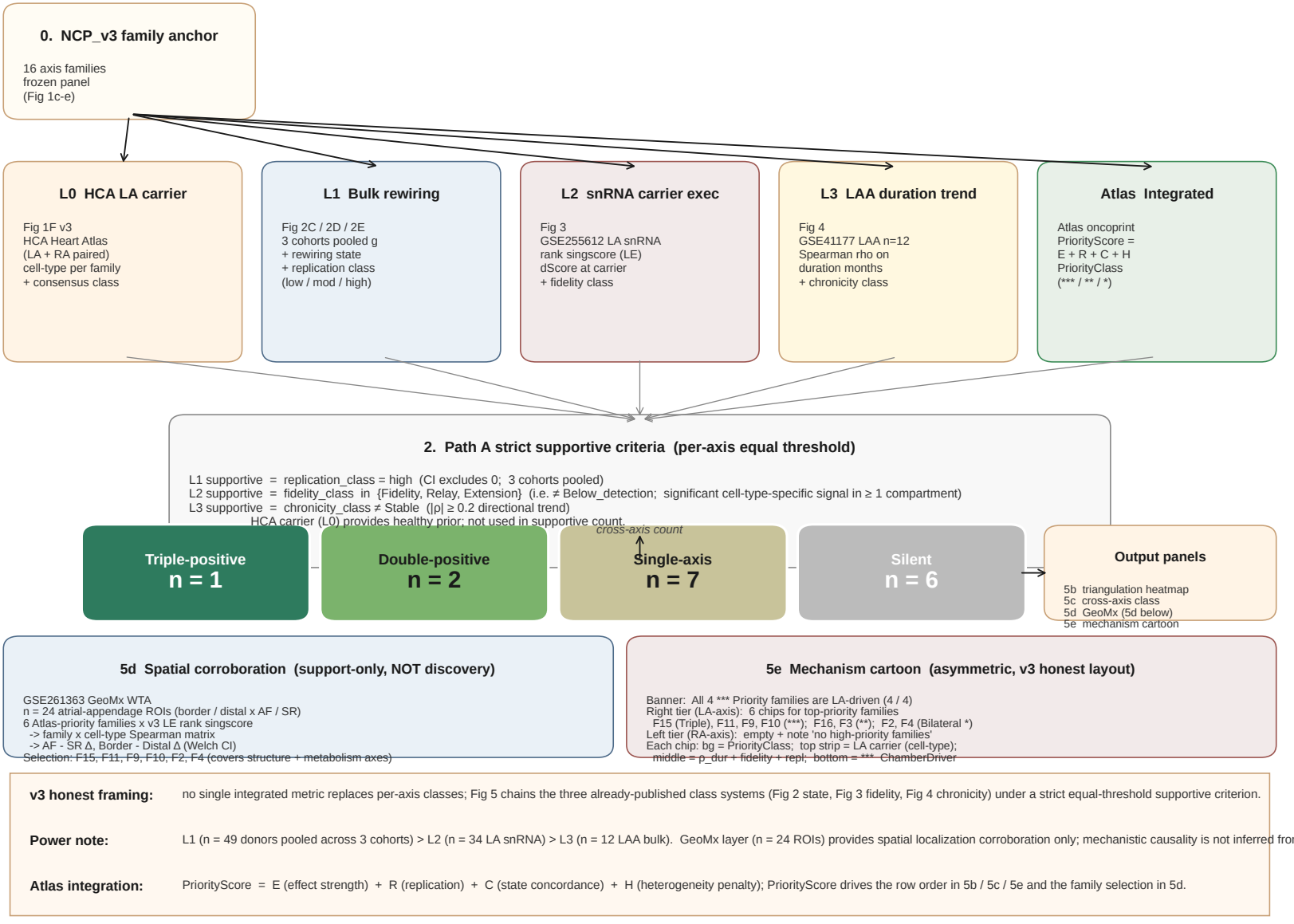
Fig 5a Cross-figure synthesis pipeline (NCP_v3, 16 families)

Integrates Fig 1F (carrier prior), Fig 2 (bulk rewiring), Fig 3 (snRNA carrier exec) and Fig 4 (duration trend) under a strict per-axis supportive criterion (Path A).



v3 honest framing: no single integrated metric replaces per-axis classes; Fig 5 chains the three already-published class systems (Fig 2 state, Fig 3 fidelity, Fig 4 chronicity) under a strict equal-threshold supportive criterion.

Power note: L1 (n = 49 donors pooled across 3 cohorts) > L2 (n = 34 LA snRNA) > L3 (n = 12 LAA bulk). GeoMx layer (n = 24 ROIs) provides spatial localization corroboration only; mechanistic causality is not inferred from

Atlas integration: PriorityScore = E (effect strength) + R (replication) + C (state concordance) + H (heterogeneity penalty); PriorityScore drives the row order in 5b / 5c / 5e and the family selection in 5d.